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Money, Income, Prices, and Interest Rates

By BENJAMIN M. FRIEDMAN AND KENNETH N. KUTTNER*

Including data from the 1980's sharply weakens the postwar time-series evidence indicating significant relationships between money (however defined) and nominal income or between money and either real income or prices separately. Focusing on data from 1970 onward destroys this evidence altogether. Evidence indicating cointegration of real income and real money balances, with due allowance for the effect of interest rates, also deteriorates when the sample extends through the 1980's. A positive finding is that the spread between the commercial paper rate and the Treasury bill rate consistently contains highly significant information about future movements in real income. (*JEL* E52)

Economists have long understood that the quantity of money, or its growth rate, can play a useful role in the monetary policy process only to the extent that fluctuations in money over time regularly and reliably correspond to fluctuations in income, prices, or whatever other aspects of economic activity the central bank seeks to influence. The same is true, of course, for any other financial quantity (e.g., nonmoney assets, measures of credit) or, for that matter, interest rates and any other financial prices. Especially in the case of money, a rich literature developed over many years has investigated in some detail the requirements that the relationships connecting money to income or prices must satisfy in order to warrant focusing monetary policy on money in any of several specific ways (see e.g., the literature surveyed in Friedman [1990]). An equally rich empirical literature has repeatedly sought to establish whether these requirements have actually been satisfied at specific times and in specific places (see e.g., the literature surveyed by Bennett T. McCallum [1985]).

Different ways of conceptually basing the monetary policy process on money place different empirical requirements on the relationships between money and the economic variables that are of ultimate policy concern. These relationships in turn depend on such basic dimensions of economic behavior as the nature of the economy's aggregate supply process, the degree of price flexibility, the interest and wealth elasticities of aggregate demand, and importantly, the public's money-demand behavior and the banks' money-supply behavior. For example, under familiar circumstances using money as an "intermediate target" (i.e., determining the money growth rate most likely *ex ante* to be consistent with the central bank's macroeconomic policy objectives and then conducting monetary policy operations during some time interval as if achieving money growth along that path were itself the policy objective) is closer to being optimal as the demand for money is closer to being both nonstochastic and interest-inelastic. Looser relationships suffice for it to be optimal to use money as an "information variable," that is, adjusting policy operations during some interval in response to actual money growth that departs from the *ex ante* path and perhaps also in response to analogous departures for other variables, but in any case not in a way designed

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necessarily to restore money growth to the *ex ante* path.[1]

However, what is essential to either of these ways of proceeding—and to others besides—is that there be at least some reliably exploitable connection between money and either income or prices, so that observed departures of money from some *ex ante* path bear a systematic implication for income or prices in the future. Otherwise, as a variable that the central bank cannot set directly as a policy instrument, money has no role in the policy process. From an information-variable perspective, there is no point to the central bank's reacting to fluctuations in money if those fluctuations bear no implication for subsequent movements in income or prices. From an intermediate-target perspective, there is even less point to making policy as if controlling money were stochastically equivalent to controlling income and prices if in fact there is no relation between them.

It is for just this reason that the events of the 1980's have proved so subversive to what had almost come to be standard ways of thinking about monetary policy, not just in the United States but in many other countries as well. While there was never any lack of debate about the strength or weakness of the empirical relationships connecting money to income and prices, and therefore about the appropriate role of money in the monetary policy process, before the 1980's there was widespread agreement that fluctuations in money did contain at least potentially useful information about future income and price movements. In the 1980's, however, the empirical basis underlying that agreement disappeared. Empirical investigation based on sample periods that include the 1980's simply does not lead to results corroborating what were commonly accepted as facts of economic behavior not so many years earlier. It is not surprising that in this environment some central banks, including the Federal Reserve System in the United States, have altered or abandoned the ways in which they had previously relied on money to make policy.

The object of this paper is to show how the passage of time—in particular, the experience since 1980—has altered familiar empirical relationships previously taken to support a central role for money in the monetary policy process. Section I reports results for a variety of significance tests and forecast error variance decompositions, in each case seeking to establish whether fluctuations in money or interest rates are useful for predicting subsequent fluctuations in income or prices. Here the consistent finding is that the positive results for the monetary aggregates in this context that are familiar from earlier time periods do not hold up when the sample is extended to include data from the 1980's. A further consistent result is that the difference between the commercial paper rate and the Treasury bill rate does contain incremental information about real income (but not prices). Section II focuses in more detail on this predictive power of the paper–bill spread. Section III shifts the paper's focus to long-run relationships, reporting results for tests of the cointegration of movements of money and income. Here, as in Section I, the relationships that would have to hold in order to warrant using money as the central focus of monetary policy disappear when the analysis includes data from the 1980's. Section IV briefly concludes by drawing together the implications of these findings for monetary policy.

I. Autoregression Tests and Variance Decompositions

Ever since the early work of Christopher A. Sims (1972), empirical consideration of whether money (or any other aggregate) can usefully play a role in the monetary policy process has appropriately focused not just on whether fluctuations of money help predict future fluctuations of income (or prices, etc.), but on whether they help predict future fluctuations of income that are not already predictable on the basis of fluctua-

[1]John H. Kareken et al. (1973) first formally introduced the information-variable concept into the analysis of monetary policy. See Friedman (1975) for a formal analysis of the intermediate-target-variable procedure.

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TABLE 1—*F* STATISTICS FOR FINANCIAL VARIABLES IN NOMINAL-INCOME EQUATIONS

Variable	1960:2–1979:3	1960:2–1990:4	1970:3–1990:4
A. *Three-Variable System (Nominal Income, Fiscal Variable, Financial Variable):*			
Δ ln(base)	3.68**	1.85	0.82
Δ ln(M1)	7.23**	3.75**	2.27†
Δ ln(M2)	5.12**	4.49**	1.85
Δ ln(credit)	5.54**	1.55	0.21
Δr_P	1.60	5.55**	4.39**
Δr_B	1.81	4.82**	3.96**
$r_P - r_B$	4.43**	4.68**	3.16*
B. *Two-Variable System (Nominal Income, Financial Variable):*			
Δ ln(base)	4.07**	2.19†	0.93
Δ ln(M1)	7.39**	3.75**	2.14†
Δ ln(M2)	5.19**	5.15**	2.37†
Δ ln(credit)	4.96**	1.68	0.39
Δr_P	1.91	5.84**	4.36**
Δr_B	2.15†	5.05**	3.83**
$r_P - r_B$	5.34**	5.38**	3.35*

Notes: Estimated regressions use four lags of each variable. Income is GNP in current dollars, and the fiscal variable is mid-expansion federal expenditures.

†Statistically significant at the $P \leq 0.1$ level.

*Statistically significant at the $P \leq 0.05$ level; **Statistically significant at the $P \leq 0.01$ level.

tions of income itself or other readily observable variables. Especially in the context of the information-variable approach to monetary policy, this interpretation of Sims-type autoregression tests is precisely what is at issue, and the much-debated question of whether such tests constitute valid tests of "causality" is beside the point. As long as movements in money do contain information about future movements in income beyond what is already contained in movements in income itself, monetary policy can exploit that information by responding to observed money growth regardless of whether the information it contains reflects true causation, reverse causation based on anticipations, or mutual causation by some independent but unobserved influence.

The first four lines in the upper portion of Table 1 present *F* statistics for tests, across different time periods, of the null hypothesis that all of the coefficients on the lagged growth of either the monetary base, M1, M2, or credit (that is, all of the β_i) are zero in autoregressions of the form

$$(1)\quad \Delta y = \alpha + \sum_{i=1}^{4} \beta_i \Delta m_{t-i} + \sum_{i=1}^{4} \gamma_i \Delta g_{t-i} + \sum_{i=1}^{4} \delta_i \Delta y_{t-i} + u_t$$

where y, m, and g (all in natural logarithms) are, respectively, nominal income, the financial variable indicated, and mid-expansion federal expenditures; α, the β_i, the γ_i, and the δ_i are coefficients to be estimated; and u is a disturbance term.[2]

[2]The monetary base and the two "M's" are based on the conventional Federal Reserve Board definitions. Credit is the outstanding indebtedness of domestic nonfinancial borrowers. Income is gross national product. (Substituting domestic absorption for gross national product does not systematically change the results reported here or in most of the tables introduced below.) All data except interest rates are seasonally adjusted.

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The lower portion of the table shows F statistics based on analogous equations excluding the government-spending variable. The results reported are based on standard quarterly data for three sample periods: 1960:2–1979:3, 1960:2–1990:4, and 1970:3–1990:4.

For the sample spanning 1960:2–1979:3 (i.e., from the earliest time for which the Federal Reserve provides data corresponding to its current definitions of the monetary aggregates until the introduction of its new monetary policy procedures in October 1979), the F statistics in the first column of Table 1 show that the base, M1, M2, and credit each contained information about future income movements that was statistically significant at the 0.01 level, regardless of whether the analysis includes the fiscal variable. Extending the sample to include data through year-end 1990 eliminates this significance for the base and for credit. Also, beginning the sample in 1970:3 (i.e., after the removal of the Regulation Q ceiling on interest paid on large certificates of deposit and the beginning of the [sporadic] targeting of money growth for purposes of monetary policy) leaves only M1 significant among these three aggregates, even at the 0.10 level in the presence of the fiscal variable, and leaves only M1 and M2 significant (again, only at the 0.10 level) without the fiscal variable.[3]

For purposes of comparison, Table 1 also presents F statistics for analogous autoregressions in which variable Δm in (1) is replaced by, successively, the interest rate on prime 4–6 month commercial paper, the 90-day Treasury bill rate, and the difference between these two interest rates.[4] Introducing an interest rate in relationships like those under study here, either in place of or in addition to a financial aggregate, has also been familiar since the work of Sims (1980). Sims (1980) and Friedman (1983) both used the commercial paper rate for this purpose. Robert B. Litterman and Lawrence Weiss (1985), Martin Eichenbaum and Kenneth J. Singleton (1986), and James H. Stock and Mark W. Watson (1989a) all used the Treasury bill rate.

These relationships too have changed with the passage of time, although in the case of the two interest rates the changes have been at least partly in the direction of stronger, rather than weaker, ability to predict income. For the sample ending in 1979, neither the commercial paper rate nor the Treasury bill rate contained statistically significant information about future fluctuations in income. By contrast, the spread between the two rates contained information about income that was significant at the 0.01 level. For both the 1960–1990 and the 1970–1990 samples, however, the information contained in each of the three interest rate variables is significant at least at the 0.05 level.

Tables 2 and 3 present analogous results based on equations in which the variable whose movements are to be explained is, in turn, real income and then the price level, and the equation includes lagged values of both real income and the price level as regressors.[5] As is consistent with much of

[3]Restricting the sample to the period since the Federal Reserve's change in operating procedures (i.e., 1979:4–1990:4), renders all four aggregates not significant, even at the 0.10 level. Including a dummy variable for the 1980's in the regressions based on the longer samples (a suggestion made by an anonymous referee) makes only a small difference for the results reported above. In the 1970–1990 equations for M1, for example, adding a dummy variable equal to 1 beginning in 1979:4 results in F statistics of 2.45 when the fiscal variable is included and 2.26 when it is excluded; both are again significant only at the 0.10 level. The F statistics in the corresponding M2 equations are 1.29 and 1.62, respectively. Neither of these values is significant at the 0.10 level.

[4]We are grateful to Bennett McCallum for suggesting the inclusion of interest rates in this analysis for purposes of comparison. Both interest rates are entered in the regressions in difference form, while the spread is entered in level form, because univariate stationarity tests indicate that the two interest rates (like income and the financial aggregates) are each difference-stationary, while the spread is stationary in levels.

[5]Real income is gross national product in 1982 dollars. The price level is the corresponding deflator.

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TABLE 2—*F* STATISTICS FOR FINANCIAL VARIABLES IN REAL-INCOME EQUATIONS

Variable	1960:2–1979:3	1960:2–1990:4	1970:3–1990:4
A. *Four-Variable System (Real Income, Price Index, Fiscal Variable, Financial Variable):*			
Δ ln(base)	1.22	0.77	0.52
Δ ln(M1)	2.71*	2.85*	1.81
Δ ln(M2)	3.63**	4.32**	1.81
Δ ln(credit)	2.34†	1.02	0.16
Δr_P	1.55	5.61**	3.94**
Δr_B	1.81	4.52**	3.44*
$r_P - r_B$	7.10**	7.16**	4.68**
B. *Three-Variable System (Real Income, Price Index, Financial Variable):*			
Δ ln(base)	1.27	0.91	0.98
Δ ln(M1)	2.59*	2.65*	1.77
Δ ln(M2)	3.78**	4.66**	2.19†
Δ ln(credit)	1.97	1.21	0.34
Δr_P	1.95	5.80**	4.14**
Δr_B	2.21†	4.76**	3.62**
$r_P - r_B$	8.12**	7.70**	5.32**

Notes: Estimated regressions use four lags of each variable. Income is GNP in 1982 dollars, the price index is the implicit GNP deflator, and the fiscal variable is mid-expansion federal expenditures.

†Statistically significant at the $P \leq 0.1$ level.

*Statistically significant at the $P \leq 0.05$ level; **Statistically significant at the $P \leq 0.01$ level.

the existing literature, the pre-1980 evidence for the three aggregates is mixed, depending on which one the equation includes and on whether the dependent variable is income or prices.[6] Including data through 1990 leaves intact the positive results for M1 and M2 in the equations for real income but eliminates altogether the significance of any of the financial aggregates in the equations for prices. Beginning the sample in 1970 eliminates the significance of the four aggregates in the real-income equations as well, except for just one *F* statistic that remains significant at the 0.10 level (and, of course, 1 out of 10 would be expected in the context of no systematic relationship whatsoever).[7]

The *F* statistics reported for the three interest rate variables in Tables 2 and 3—especially Table 2—again tell a quite different story. In the 1960–1979 sample, neither interest rate contained much if any information about future movements in real income. For the 1960–1990 sample, however, this relationship is significant at the 0.01 level for either interest rate, and it remains so (except for the drop to the 0.05 level for the bill rate in the presence of the fiscal variable) when the observations from

[6]The sensitivity of the results of such tests to these and other aspects of the underlying specification are well known. See, for example, Arnold Zellner (1985).

[7]In these equations as well, including a dummy variable for the 1980's (see again footnote 3) makes little difference. In the only case (out of 16) in which the aggregate is significant at the 0.10 level, in the equation for either real income or prices over the 1970–1990 sample (M2 in the real-income equation, when the fiscal variable is omitted), adding the dummy variable reduces the *F* statistic from 2.19, as reported in Table 2, to 1.81, which falls below the 0.10 critical value. Adding the dummy variable increases the *F* statistic on the monetary base in the equation for prices from 1.96, as reported in Table 3, to 2.04, which is significant at the 0.10 level. None of the other significance levels reported in Tables 2 or 3 changes on addition of the dummy variable.

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TABLE 3—*F* STATISTICS FOR FINANCIAL VARIABLES IN PRICE EQUATIONS

Variable	1960:2–1979:3	1960:2–1990:4	1970:3–1990:4
A. *Four-Variable System (Real Income, Price Index, Fiscal Variable, Financial Variable):*			
Δ ln(base)	5.13**	1.85	1.68
Δ ln(M1)	3.96**	0.83	0.54
Δ ln(M2)	1.61	1.08	0.92
Δ ln(credit)	2.88*	0.65	0.13
Δr_P	2.42†	2.43†	2.39†
Δr_B	2.42†	2.11†	1.63
$r_P - r_B$	1.49	1.72	1.74
B. *Three-Variable System (Real Income, Price Index, Financial Variable):*			
Δ ln(base)	1.85	5.53**	1.96
Δ ln(M1)	0.92	4.12**	0.60
Δ ln(M2)	1.20	1.85	0.98
Δ ln(credit)	0.72	2.96*	0.08
Δr_P	2.36†	2.52*	2.55*
Δr_B	2.06†	2.66*	1.81
$r_P - r_B$	1.78	1.59	1.79

Notes: Estimated regressions use four lags of each variable. Income is GNP in 1982 dollars, the price index is the implicit GNP deflator, and the fiscal variable is mid-expansion federal expenditures.

†Statistically significant at the $P \leq 0.1$ level.

*Statistically significant at the $P \leq 0.05$ level; **Statistically significant at the $P \leq 0.01$ level.

the 1960's are dropped. By contrast, whatever information these interest rates contained about the price level in the 1960–1979 sample is less apparent in either sample that includes the 1980's.

Perhaps the single most striking result in the entire analysis so far is the strong relationship between real income (though not the price level) and the interest-rate-spread variable. The difference between the commercial paper rate and the Treasury bill rate contains information about future fluctuations in real income that is significant at the 0.01 level, either with or without the fiscal variable, in each of the three samples.

Statistical significance, of course, is not the only metric for judging dynamic relationships among two or more time series. Following Sims (1980), an important criterion emphasized in at least some of the empirical literature of the money–income relationship is the ability of money (or other variables) to account for the forecast error variance of income at horizons of interest in a business-cycle context. Table 4A reports variance decompositions for nominal income, at both four- and eight-quarter horizons, generated by a series of three-variable (nominal income, fiscal variable, and financial variable) vector autoregressions in which the financial variable included is, successively, each of those considered in Tables 1–3. Within each autoregression system, the right-hand side of each equation is identical to that given in (1).

The sharp deterioration in the money–income relationship is readily apparent by this alternative criterion as well. In the 1960–1979 sample, M1, M2, and credit each account for 20–30 percent of the income error variance, at either four or eight quarters, and the monetary base share is in the 10–15-percent range. Each of these shares is statistically significant.[8] In the 1970–1990

[8]The ranges given in the tables indicate 95-percent confidence intervals computed from Monte Carlo simulations.

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TABLE 4—DECOMPOSITION OF NOMINAL AND REAL-INCOME VARIANCE FOR ALTERNATIVE FINANCIAL VARIABLES

Variables	1960:2–1979:3		1970:3–1990:4	
	4-quarter horizon	8-quarter horizon	4-quarter horizon	8-quarter horizon
A. *Three-Variable System with Nominal Income:*				
Δ ln(base)	13±9	15±8	6±8	8±10
Δ ln(M1)	27±12	27±12	10±9	11±9
Δ ln(M2)	20±12	22±14	11±10	14±12
Δ ln(credit)	23±13	25±13	3±5	4±5
Δr_P	7±8	10±10	16±10	17±11
Δr_B	5±6	10±10	15±10	15±10
$r_P - r_B$	18±12	19±11	15±11	16±11
B. *Four-Variable System with Real Income:*				
Δ ln(base)	4±5	6±5	3±4	5±5
Δ ln(M1)	12±10	15±11	8±7	9±8
Δ ln(M2)	11±8	14±10	9±9	11±9
Δ ln(credit)	7±8	9±8	3±4	3±4
Δr_P	8±6	11±7	16±10	17±10
Δr_B	5±6	11±9	13±10	15±10
$r_P - r_B$	27±12	27±11	21±11	20±11

Notes: Variance decompositions are computed via Monte Carlo simulations with 500 draws. Ranges indicated represent approximate 95-percent confidence intervals. The orthogonalization order is: nominal income, fiscal variable, financial variable (in part A); real income, price index, fiscal variable, financial variable (in part B). (See also notes to Tables 1 and 2.)

sample, however, the respective shares accounted for by the base, M1, and M2 all drop to about one-half what they were earlier, and the credit share almost disappears. In terms of comparisons to the interest rate variables, in the earlier sample M1, M2, and credit outperform all three interest rate variables (including the paper–bill spread), while in the later sample each of the three interest rate variables outperforms all four aggregates.

Table 4B presents analogous forecast error variance decompositions for real income based on a set of four-variable (real income, prices, fiscal variable, and financial variable) vector autoregressions. Here the basic story is similar, in that the share of real-income variance accounted for by each of the four aggregates deteriorates from the earlier to the later sample. The most striking difference between these results and those shown above is that even in the 1960–1979 sample the respective *real*-income variance shares corresponding to the various aggregates were never very great anyway.[9] By contrast, the share of real-income variance accounted for by the paper–bill spread easily dominates any of the other six financial variables considered, for either sample period.

Finally, results (not shown here) analogous to those reported in Tables 1–4 but based on monthly data,[10] with industrial

[9]The difference in this respect between the results shown in Table 4 and those shown in Table 2 corresponds to an important but often ignored distinction in the existing literature. Some authors (e.g., Stock and Watson, 1989a) have focused on significance in autoregressions, while others (e.g., Sims, 1980) have focused on forecast error variance decompositions, with the result that different researchers have appeared to contradict one another even for work based on the same sample period.

[10]There is a large literature testing the money–income relationship using monthly data (e.g., Sims, 1980; Eichenbaum and Singleton, 1986; Stock and Watson, 1989a).

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production used in place of real income and with the producer price index used instead of the implicit deflator, tell approximately the same story.[11] In equations for nominal income (industrial production times the price index) all four aggregates are significant at the 0.01 level in the 1960.2–1979.9 sample, but only the base and M2 are significant, and those merely at the 0.10 level, in comparable equations for 1970.7–1990.12. Both M1 and M2 are significant at the 0.01 level (and credit at the 0.10 level) in equations for industrial production over 1960–1979 but of these only M2 survives (at the 0.05 level) for 1970–1990.[12] The monetary base is significant at the 0.01 level and credit is significant at the 0.10 level in equations for the price index over 1960–1979, but only the base survives (and that merely at the 0.10 level) for 1970–1990. Variance-decomposition results based on corresponding vector autoregression systems estimated with monthly data likewise show a sharp deterioration in the share of either the nominal- or the real-income forecast error accounted for by any of the four aggregates. By contrast, both significance tests and variance decompositions show that the relationship of either nominal or real income to the three interest rate variables considered above, and especially the relationship of real income to the paper–bill spread, exhibits no such deterioration.

II. Real Income and the Paper–Bill Spread

The significance tests and variance decompositions shown in Tables 1–4 reinforce earlier literature in suggesting that market interest rates contain statistically significant information about future fluctuations in income—especially real income—and the strong performance of the paper–bill spread in those tables stands out in particular. Tables 5–9 further confirm these findings by presenting comparable results based on real-income equations that include a financial aggregate together with the commercial paper rate, the Treasury bill rate, or the paper–bill spread.

Table 5 shows that, for the most part, neither the commercial paper rate nor any of the financial aggregates was particularly significant (in the presence of one another) in equations for real income during the 1960–1979 sample. In the two samples that include the 1980's, however, the paper rate is significant at the 0.05 level or better in the presence of any of the aggregates but M2. Analogous results for the Treasury bill rate, shown in Table 6, are roughly similar, except that here the interest rate is significant in the early sample as well. In the presence of the bill rate, both M1 and M2 are significant at the 0.01 level, and credit is significant at the 0.10 level in the 1960–1979 sample, but none of these results for the aggregates survives the inclusion of data from the 1980's.

The F statistics reported in Table 7 show that the paper–bill spread consistently dominates any of the financial aggregates in anticipating fluctuations in real income, regardless of sample period. Neither the monetary base nor M1 nor M2 is ever significant in the presence of the spread variable in any of the three samples, and credit is significant in the presence of the spread only in the 1960–1979 sample. By contrast, the spread is always significant at the 0.05 level or better in the presence of any of the financial aggregates. (Except when the aggregate is M2, the spread is always significant at the 0.01 level.)

Table 8 presents forecast error variance decompositions that further reconfirm this strong performance of the paper–bill spread. With only one exception, the spread dominates any of the financial aggregates in accounting for the variance of real income, regardless of the sample period and regardless of whether the ordering of the variables

[11] These monthly results are analogous to those shown in the lower panels of Tables 1–3, in that they omit the fiscal variable. Data for mid-expansion federal expenditures are not available on a monthly basis.

[12] In monthly equations for industrial production specified as recommended by Stock and Watson (1989a), that is, including an interest rate term and a time trend along with the financial aggregate and with lag length chosen to *favor* the significance of money, none of the aggregates is significant over the 1970.7–1990.12 sample, even at the 0.10 level.

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TABLE 5—*F* STATISTICS FOR FINANCIAL AGGREGATES AND THE COMMERCIAL PAPER RATE IN REAL-INCOME EQUATIONS

Equation	1960:2–1979:3	1960:2–1990:4	1970:3–1990:4
Equation with Monetary Base:			
Excluding Δ ln(base)	1.32	0.72	1.58
Excluding Δr_P	1.97	5.40**	4.67**
Equation with M1:			
Excluding Δ ln(M1)	2.05†	0.91	0.75
Excluding Δr_P	1.46	3.76**	2.87*
Equation with M2:			
Excluding Δ ln(M2)	2.35†	1.14	0.19
Excluding Δr_P	0.74	2.11	1.84
Equation with Credit:			
Excluding Δ ln(credit)	2.37†	0.75	0.54
Excluding Δr_P	2.35†	5.10**	4.20**

†Statistically significant at the $P \leq 0.1$ level.
*Statistically significant at the $P \leq 0.05$ level; **Statistically significant at the $P \leq 0.01$ level.

TABLE 6—*F* STATISTICS FOR FINANCIAL AGGREGATES AND THE TREASURY BILL RATE IN REAL-INCOME EQUATIONS

Equation	1960:2–1979:3	1960:2–1990:4	1970:3–1990:4
Equation with Monetary Base:			
Excluding Δ ln(base)	1.48	0.52	1.42
Excluding Δr_B	2.39†	4.18**	3.97**
Equation with M1:			
Excluding Δ ln(M1)	3.60**	1.79	1.30
Excluding Δr_B	3.22*	3.77**	3.01*
Equation with M2:			
Excluding Δ ln(M2)	4.61**	1.95	0.42
Excluding Δr_B	3.05*	2.04†	1.64
Equation with Credit:			
Excluding Δ ln(credit)	1.81†	0.71	0.41
Excluding Δr_B	2.04†	4.07**	3.54*

†Statistically significant at the $P \leq 0.1$ level.
*Statistically significant at the $P \leq 0.05$ level; **Statistically significant at the $P \leq 0.01$ level.

for purposes of orthogonalization is income-prices-aggregate-spread or income-prices-spread-aggregate. The sole exception is the case of M2 for the 1960–1979 sample, in which the aggregate accounts for a modestly greater share of real-income variance if it is ordered before the spread. Reversing the ordering, however, causes the comparison to go much more decisively in the other direction.

A potential role for market interest rates in providing information about future in-

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TABLE 7—*F* STATISTICS FOR FINANCIAL AGGREGATES AND THE PAPER–BILL SPREAD IN REAL-INCOME EQUATIONS

Equation	1960:2–1979:3	1960:2–1990:4	1970:3–1990:4
Equation with Monetary Base:			
Excluding Δ ln(base)	1.49	0.90	1.54
Excluding $r_P - r_B$	8.04**	7.45**	5.66**
Equation with M1:			
Excluding Δ ln(M1)	1.63	1.59	1.09
Excluding $r_P - r_B$	6.58**	6.29**	4.32**
Equation with M2:			
Excluding Δ ln(M2)	0.36	0.75	0.19
Excluding $r_P - r_B$	3.75**	3.33*	2.83*
Equation with Credit:			
Excluding Δ ln(credit)	2.73*	0.92	0.47
Excluding $r_P - r_B$	8.79**	7.12**	5.24**

*Statistically significant at the $P \leq 0.05$ level; **Statistically significant at the $P \leq 0.01$ level.

come is hardly surprising a priori. Money growth, after all, is not the only potential source of information about the state of financial markets, nor the only potential measure of whatever effects financial phenomena may be exerting on real economic activity. It makes sense, therefore, to broaden the class of financial variables considered to be potentially of use in this context to include not just financial quantity variables, like money or credit, but also financial price variables including interest rates. The results shown in Tables 7 and 8 further indicate that *relationships between different interest rates* can also be relevant in this context.

The respective interest rates on Treasury bills (unsecured short-term discount obligations of the U.S. government) and commercial paper (analogous unsecured borrowings of corporations) may vary relative to one another for any of a variety of reasons. Most obviously, commercial paper is subject to private default risk, while Treasury bills are backed by the full faith and credit of the U.S. government. Interest earned on both instruments is identically taxed at the federal level, but interest from U.S. Treasury obligations is exempt from individual income taxes and corporate profit taxes imposed at the state or municipal level.[13] Treasury bills also provide greater liquidity than commercial paper in several market contexts. Each of these three distinctions gives rise to sources of imperfect covariation between the two interest rates, in ways that may plausibly be related to fluctuations in real income. Investors' assessment of private default risk, the identity and hence tax rate of the marginal investor, and the value placed by investors on the superior liquidity of bills may all change at least in part systematically over the business cycle.

In addition, to the extent that these distinguishing features (and perhaps others as well) render commercial paper and Treasury bills imperfect portfolio substitutes, changes in the relative supplies of the two instruments will also affect their relative returns. Here too, some of these influences may vary systematically over the business

[13]Interest on Treasury securities is subject to state-level business franchise taxes, as well as to excise taxes imposed on bank profits; see Friedman and Kuttner (1992).

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TABLE 8—DECOMPOSITION OF REAL-INCOME VARIANCE FOR FINANCIAL AGGREGATES AND THE PAPER–BILL SPREAD

System	1960:2–1979:3		1970:3–1990:4	
	Aggregate	Spread	Aggregate	Spread
System with Monetary Base:				
Ordering = aggregate-spread				
Variance at 4 quarters	3±4	30±12	4±5	26±13
Variance at 8 quarters	6±6	30±12	5±6	25±13
Ordering = spread-aggregate				
Variance at 4 quarters	3±4	30±13	3±4	25±13
Variance at 8 quarters	6±5	30±14	5±5	25±13
System with M1:				
Ordering = aggregate-spread				
Variance at 4 quarters	15±11	23±12	8±8	22±13
Variance at 8 quarters	18±11	23±12	10±9	21±13
Ordering = spread-aggregate				
Variance at 4 quarters	5±6	32±13	5±5	25±14
Variance at 8 quarters	9±7	32±13	6±6	25±14
System with M2:				
Ordering = aggregate-spread				
Variance at 4 quarters	16±11	12±9	9±9	14±11
Variance at 8 quarters	19±11	13±8	11±10	15±10
Ordering = spread-aggregate				
Variance at 4 quarters	5±9	23±12	2±3	22±12
Variance at 8 quarters	9±9	22±12	5±7	21±12
System with Credit:				
Ordering = aggregate-spread				
Variance at 4 quarters	6±6	31±13	5±6	23±13
Variance at 8 quarters	8±7	31±14	6±6	23±13
Ordering = spread-aggregate				
Variance at 4 quarters	6±7	31±14	4±5	24±12
Variance at 8 quarters	8±7	31±13	5±6	24±12

Notes: Variance decompositions are computed via Monte Carlo simulations with 500 draws. Ranges indicated represent approximate 95-percent confidence intervals. The four-variable VAR includes four lags of each of the following: real income, price index, financial aggregate, and paper–bill spread.

cycle, as a widening or narrowing government deficit alters the Treasury's borrowing program, changing business cash flows have an analogous effect on corporate borrowing, and the stance of monetary policy in part determines how much of corporations' borrowing banks (as opposed to the commercial-paper market) can accommodate.

Table 9 presents the results of an even more rigorous series of tests investigating whether the paper–bill spread contains information about future fluctuations in real income. The table shows F statistics, for the usual three sample periods, based on autoregressions of the form

$$(2)\quad \Delta y_t = \alpha + \sum_{i=1}^{4} \beta_i \Delta m_{t-i} + \sum_{i=1}^{4} \gamma_i \Delta r_{P,t-i} + \sum_{i=1}^{5} \delta_i (r_P - r_B)_{t-i} + \sum_{i=1}^{4} \theta_i \Delta y_{t-i} + \sum_{i=1}^{4} \phi_i \Delta p_{t-i} + u_t$$

where y and p are the logarithms of real gross national product and the corresponding implicit deflator, respectively; m is, in

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TABLE 9— F STATISTICS FOR EXPANDED REAL-INCOME EQUATIONS

Equation	1960:2–1979:3	1960:2–1990:4	1970:3–1990:4
Equation with Monetary Base:			
Excluding Δ ln(base)	1.61	0.75	1.61
Excluding Δr_P	1.23	2.12†	1.99
Δr_B constraint	4.95**	4.03**	2.75*
Excluding $r_P - r_B$	5.34**	3.10*	2.26†
Equation with M1:			
Excluding Δ ln(M1)	3.67**	1.75	1.57
Excluding Δr_P	2.92*	2.34†	2.07†
Δr_B constraint	5.03**	3.84**	2.81*
Excluding $r_P - r_B$	6.73**	3.84**	2.92*
Equation with M2:			
Excluding Δ ln(M2)	1.99	0.53	0.13
Excluding Δr_P	2.47†	1.99	1.63
Δr_B constraint	2.74*	2.62*	2.24†
Excluding $r_P - r_B$	4.76**	2.56*	2.09†
Equation with Credit:			
Excluding Δ ln(credit)	1.91	1.11	0.67
Excluding Δr_P	0.46	2.41†	1.85
Δr_B constraint	4.97**	4.19**	2.76*
Excluding $r_P - r_B$	4.65**	3.40**	2.28†

†Statistically significant at the $P \leq 0.1$ level.

*Statistically significant at the $P \leq 0.05$ level; **Statistically significant at the $P \leq 0.01$ level.

turn, the logarithm of each of the four financial aggregates considered above; r_P is the commercial paper rate; r_B is the Treasury bill rate; and u is again a disturbance term.[14] The main point is that (2) includes both an aggregate and the interest rate along with the paper–bill spread.

The first F statistic reported in each panel of Table 9, for the null hypothesis that the β coefficients are uniformly zero, indicates that with the sole exception of M1 in the 1960–1979 sample, none of the four aggregates contains information about real income that is statistically significant in this expanded context, even at the 0.10 level, in any of the three samples. The second F statistic reported in each panel, testing the null hypothesis that the γ coefficients are all zero, indicates that the commercial paper rate is significant only sporadically in the presence of the aggregate and the paper–bill spread.

The third F statistic reported in each panel of Table 9 is for a test of the null hypothesis that the individual γ and δ coefficients in (2) just cancel one another so that the correct specification simply includes the bill rate while excluding the paper rate.[15] With only one exception (in which the constraint is still rejected at the 0.10 level), the results warrant rejecting this constraint at the 0.05 level in each of the three

[14]Entering the interest rates as differences and the spread as a level again reflects these series' univariate stationarity properties; see footnote 4.

[15]Given that the paper rate and the paper–bill spread enter (2) in differences and in levels, respectively, the constraint being tested is

$$\delta_1 + \gamma_1 = 0$$

$$\delta_2 + \gamma_2 - \gamma_1 = 0$$

$$\vdots$$

$$\delta_5 - \gamma_4 = 0.$$

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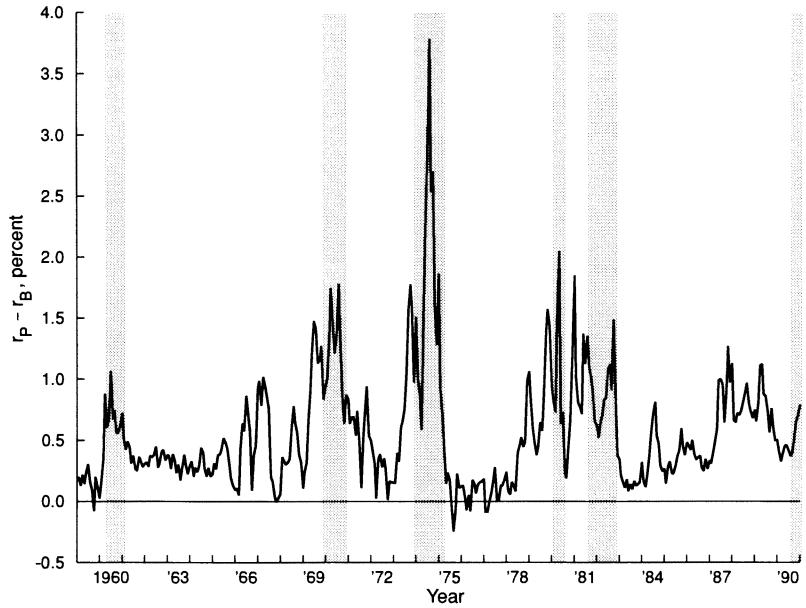


FIGURE 1. PAPER–BILL SPREAD, 1959–1990

Note: Shaded areas denote recessions.

samples and in the presence of any of the four aggregates. In other words, what contains statistically significant information about subsequent fluctuations of real income is neither the paper rate nor the bill rate individually, but the relationship between the two.

The last F statistic reported in each panel of Table 9 is simply for the null hypothesis that all of the δ coefficients are zero. The results uniformly warrant rejecting this constraint at the 0.05 level or better in the two samples including the 1960's and at either the 0.05 level (once) or the 0.10 level (three times) for the later sample. Even in the presence of both an aggregate and an interest rate, therefore, the paper–bill spread still contains at least marginally significant information about subsequent fluctuations in real income. Together with the results reported immediately above (the rejection of the "Δr_B constraint"), these results also indicate that this finding does not merely reflect the inadvertant specification of the underlying equation with the "wrong" interest rate.

Results (not shown here) analogous to those reported in Table 7 but based on monthly data make the same point even more strongly. The exclusion restriction for the paper–bill spread is uniformly rejected at the 0.01 level, regardless of which aggregate the equation includes, even for the 1970–1990 sample. The Δr_B constraint is also rejected at the 0.01 level in all cases except that of the 1960–1979 sample when the aggregate is M2, and then it is rejected at the 0.05 level.

Figure 1 plots the monthly difference between the commercial paper rate and the Treasury bill rate, for data spanning 1959–1990, with shadings to indicate NBER-designated recessions. As the figure confirms, this spread is typically larger both during and immediately prior to recessions (although the 1990 experience, in which the spread widened much longer in advance of the recession only then to narrow again before the recession began, is an obvious counterexample). Following the discussion above, the variations in the paper–bill spread over time no doubt reflect a combi-

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nation of behavioral influences. The results reported here cannot answer the question of just which determinants of the spread account for its strong information content with respect to future real income. They do suggest, however, that the imperfect covariation between the commercial paper rate and the Treasury bill rate captures more of the relevant information about the aspects of financial markets that matter for the determination of real output than do movements in either interest rate by itself—or, for that matter, fluctuations in money.[16]

III. The Cointegration of Money and Income

The empirical tests reported in Sections I and II all focus on short-run relationships connecting the growth rate of money (or some other financial aggregate) to the growth rate of income or prices. Formulating these tests in terms of growth rates is appropriate in light of the repeated finding that money, income, prices, and interest rates are all correctly modeled as integrated series, with no tendency for the respective levels of these variables to return to a specific value or to a deterministic trend (see e.g., Charles R. Nelson and Charles I. Plosser, 1982).[17] By contrast, for some questions of potential importance in the practical conduct of monetary policy, especially those that arise in a multiyear context, what matters is the long-run relationship between the *level* of money and the *level* of income or prices.

The relationship between the respective levels of money and of income or prices is especially relevant in the context of the "base drift" problem that inevitably arises when the central bank uses as an intermediate target a money (or credit) aggregate that is endogenous over short time horizons. In particular, whenever actual money growth has differed from the growth rate targeted *ex ante*, the central bank must decide whether to "let bygones be bygones" and simply conduct monetary policy so as best to achieve the targeted growth rate thereafter or, instead, seek to return the chosen aggregate itself to the previously targeted path, thereby offsetting the initial unplanned deviation of actual from targeted money growth by a subsequent, deliberately engineered deviation in the opposite direction.[18] As Carl E. Walsh (1986) has shown, which of these two strategies is preferable depends on whether whatever disturbance to the money–income (or money–price) relationship has accompanied the original departure from the targeted path is more likely to prove permanent or transitory, and a strategy of offsetting observed deviations only in part will, in general, dominate either one. Offsetting in full all observed deviations of actual from targeted money growth is optimal only if all disturbances to the money–income relationship are transitory.

The answer to the dilemma of base drift therefore depends on whether the relationship between the respective levels of money and income (in its simplest and most familiar form, the ratio of money to income; or

[16]Some subsequent work by other researchers has likewise shown a powerful predictive role for the paper–bill spread, or something very like it. Stock and Watson's (1989b) new index of leading indicators, for example, prominently includes the paper–bill spread. Ben S. Bernanke and Alan S. Blinder (1990) found similar results using the spread between the federal-funds rates and the Treasury bill rate. See Friedman and Kuttner (1992) for an investigation of a series of hypotheses derived from considerations like those mentioned above; see also Anil K. Kashyap et al. (1991) on the potential role of monetary policy in this context.

[17]In more precise statistical terms, a series is integrated if differencing is required to render it covariance-stationary (a covariance-stationary random variable is, in turn, one whose variances and autocovariances are finite and time-invariant); see, for example, Wayne A. Fuller (1976).

[18]The origin of the label "base drift" for this problem was the complaint, often voiced during the late 1970's, that the Federal Reserve System was accommodating rising inflation by pursuing a "bygones" strategy in which the targeted rate of money growth was low each year but actual money growth on a multiyear basis was high, because with repeated upward errors, each year's *target* growth was calibrated from the higher "base" level resulting from the previous year's *actual* growth.

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its reciprocal, "velocity") is stationary. Of course, the fact that money and income are both individually nonstationary need not imply that the ratio of one to the other is also nonstationary. In statistical terms, two series are cointegrated whenever they are individually integrated yet there exists a linear combination of the two that is stationary. A stationary money:income ratio therefore means that money and income are cointegrated in logarithms, although this particular relationship would represent only one special case among the more general set of possibilities exhibiting cointegration.

For example, if the logarithms of money and income are each individually integrated but jointly obey a simple equilibrium relationship of the form

$$m = \alpha + \beta y \tag{3}$$

and the deviations of m and y from (3) are stationary, then the two variables are cointegrated (for any nonzero value of coefficient β) in that either y or m will tend to adjust, so as to restore the equilibrium relationship in (3), after any realized disturbance. (Only if $\beta = 1$, which corresponds to the familiar classical assumption of a unit income elasticity of money demand, is the money:income ratio also stationary.) Alternatively, if any disturbance producing a deviation from (3) is equally likely to increase or to decrease from each period's realized value, then y and m have no tendency to return to an equilibrium relationship and, hence, are not cointegrated.

An alternative statement of what cointegration implies that is especially useful for applications to questions about monetary policy puts the point in terms of error correction. As Robert F. Engle and C. W. J. Granger (1987) have shown, if two variables are cointegrated then it is possible to express the relationship between them as an error-correction process in which the change in either variable is a function of current and past deviations from the equilibrium value of the appropriately specified linear combination of the two variables' respective levels. For example, for income chosen as the variable that changes in response to disturbances, the equilibrium money–income relationship (3) leads to the error-correction representation

$$\Delta y_t = \kappa + A(L)\Delta y_{t-1} + B(L)\Delta m_{t-1} + \lambda(m_{t-1} - \beta y_{t-1}) + u_t \tag{4}$$

where $A(L)$ and $B(L)$ are polynomials in the lag operator, κ and λ are scalar coefficients, and u is a disturbance term, which may be serially correlated. (As the discussion below makes clear, which of y or m is the "dependent" variable in this setting is an arbitrary matter of normalization.) Subject to the lags embodied in $A(L)$ and $B(L)$, (4) implies that income will adjust fully to any persistent change in the stock of money, eventually restoring the equilibrium relationship in levels represented by (3).[19] If (3) and (4) constitute a valid description of the money–income relationship, then the level of the money stock is an appropriate intermediate target of monetary policy if the ultimate policy objective is to affect the level of income.

The form of (4), including intercept κ [thereby rendering intercept α in equilibrium relationship (3) redundant], corresponds to the maximum-likelihood procedure proposed by Søren Johansen (1992) to allow for trends in the data used in testing for cointegration while jointly estimating parameters such as β. The Johansen procedure amounts to testing the dimensionality of the space spanned by whatever cointegrating vectors exist for the set of variables in question. This is equivalent to determining the number of error-correction terms in (4), while accounting for the joint estimation of all parameters (like β) that appear in the underlying equilibrium relationships. Critical values for the resulting test statistics are given in table A2 of Johansen and Katerina Juselius (1990).[20]

[19] Indeed, (4) implies that, if y and m are cointegrated, the correctly specified vector autoregression in differences should be augmented with the stationary error-correction term $\lambda(m_{t-1} - \beta y_{t-1})$.

[20] Johansen (1992) also proposes an alternative test that is appropriate if the variables in question share a

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In the context of monetary-policy issues along the lines of the discussion of base drift, the relevant question is whether money and income are cointegrated at all, that is, whether *any* cointegrating vector exists. The relevant test is therefore of the null hypothesis that *no* cointegrating vector exists, against an alternative specifying the existence of at least one such vector. Specifying an appropriate vector of variables to test for cointegration between money and income involves several disparate issues, however.

First, the limiting distributions of the test statistics tabulated by Johansen and Juselius (1990) are based on the assumption that each of the series being tested for cointegration is individually integrated of order one. As Robert G. King et al. (1991) have shown, while evidence suggests that the logarithms of nominal income and nominal money balances are each integrated of order two (i.e., Δy and Δm are themselves nonstationary), real income and real money balances both appear to be difference-stationary. Testing for cointegration in terms of real magnitudes is therefore necessary to allow the use of established critical values.

Second, unless the demand for money is interest-insensitive or interest rates are stationary, neither of which assumption has received much support from the relevant empirical literature, money and income should not be expected to exhibit cointegration unless the cointegrating equation also includes an interest rate. Third, however, as Robert E. Lucas (1988) has pointed out in a different but parallel context, the comovement of interest rates and income during the postwar period, especially in conjunction with the increased volatility of interest rates during the last decade or so, makes estimating the income elasticity of money demand [equivalent to β in (4) above] problematic when the relationship also includes an interest rate. Lucas's solution to this problem was to include the interest rate but then simply to impose a unit income elasticity a priori, rather than to estimate jointly both income and interest elasticities.

Table 10A begins at the very simplest level by reporting values, for the usual three sample periods, of augmented Dickey-Fuller t statistics testing the cointegration among quarterly values of real income and each of the four financial aggregates based on the simple "velocity" form of the cointegrating equation

$$(m_t - p_t) - x_t = e_t \tag{5}$$

where e is a disturbance term, and lags $A(L)$ and $B(L)$ in (4) are both of length 4.[21] The test statistic warrants rejecting the null hypothesis of no cointegration at the 0.05 level in the case of M2, but not for any of the other aggregates even at the 0.10 level, over either the 1960–1979 or 1960–1990 samples. For the 1970–1990 sample, however, there is no evidence for cointegration in this form for any of the four aggregates, including M2. Apparently, data from the more "tranquil" 1960's are necessary to establish the long-run stability of "velocity" in the one case in which it appears to be stable.[22]

common trend (including the special case of no trends). In that case (4) would instead be

$$\Delta y_t = A(L)\Delta y_{t-1} + B(L)\Delta m_{t-1} + \lambda(m_{t-1} - \beta y_{t-1} - \alpha) + u_t.$$

Critical values for this version of the test appear in table A3 of Johansen and Juselius (1990). Substituting the "no-trends" version for the "with-trends" version makes no difference for the significance levels reported in Table 10.

[21] The augmented Dickey-Fuller procedure is appropriate here, rather than the Johansen procedure, because there are no parameters [like β in (3)] to be estimated in cointegrating equation (5). Hence, testing for cointegration in this form simply amounts to testing for stationarity of the "velocity" ratio.

[22] It may be tempting to explain the different results for the 1960–1990 and 1970–1990 samples by the fact that the latter is shorter, but it is slightly longer than the 1960–1979 sample for which the evidence does warrant rejecting the no-cointegration null at the 0.05 level.

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TABLE 10—TESTS FOR A STATIONARY MONEY–INCOME RELATION

Statistical test	1960:2–1979:3	1960:2–1990:4	1970:3–1990:4
A. *Univariate Tests,* $(m - p) - x = e$:			
Base	−0.09	−2.25	−2.05
M1	−1.28	−1.59	−1.89
M2	−3.51*	−3.31*	−2.38
Credit	−1.21	1.15	1.29
B. *Bivariate Tests,* $(m - p) - x - \gamma r = e$:			
Base	22.8**	12.8†	9.9
M1	21.4**	12.1	9.7
M2	12.2	14.3†	8.0
Credit	18.4*	5.8	5.3
C. *Trivariate Test,* $(m - p) - \beta x - \gamma r = e$:			
Base	25.8*	16.5	11.4
M1	25.7*	16.6	10.3
M2	29.3**	21.2†	14.4
Credit	30.8**	17.7	15.1
D. *Bivariate Tests,* $(m - p) - \beta x = e$:			
Base	5.1	8.1	6.5
M1	5.3	3.7	4.2
M2	17.2*	13.4†	7.5
Credit	17.0*	5.1	5.8

Notes: Univariate tests use the augmented Dickey-Fuller statistic. Bivariate and trivariate tests use the Johansen maximal-eigenvalue likelihood-ratio statistic. All statistics are computed using four lagged differences.

†Statistically significant at the $P \leq 0.1$ level.

*Statistically significant at the $P \leq 0.05$ level; **Statistically significant at the $P \leq 0.01$ level.

Clearly, these results represent an extreme, in that (5) allows for neither an effect due to interest rates nor a nonunit income elasticity. Table 10B reports values of Johansen's "maximal-eigenvalue" statistic testing the null hypothesis of no cointegration among real income, the real values of each of the four aggregates, and the commercial paper rate, versus the alternative of one cointegrating vector, based on the cointegrating equation

$$(6) \qquad (m_t - p_t) - x_t - \gamma r_t = e_t$$

where γ is a scalar coefficient (corresponding intuitively to the interest semielasticity of money demand), and lags $A(L)$ and $B(L)$ in (4) are again both of length 4. In each case, the alternative being tested against the null hypothesis here is that one value of γ (which may be zero) exists to render the linear combination of variables given in (6) stationary.[23] For the 1960–1979 sample, the evidence suggests, at either the 0.05 or the 0.01 level, that the monetary base, M1, and credit—but interestingly, in comparison with the results shown in the panel just above, not M2—were each individually

[23]An alternative form of the likelihood-ratio statistic, based on the sum of the eigenvalues of the relevant matrix, tests the same null hypothesis against a more general alternative. Substituting this "trace" statistic (with the alternative that the number of cointegrating vectors is one *or more*) for the maximal-eigenvalue statistic (with the alternative that the number of cointegrating vectors is 1) used here makes no material difference for the results reported.

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TABLE 11—ESTIMATES OF LONG-RUN INTEREST SEMIELASTICITY, 1960–1979

Aggregate	Interest semielasticity
Base	−0.085**
M1	−0.102**
M2	−0.009†
Credit	−0.012**

†Statistically significant at the $P \leq 0.1$ level.
**Statistically significant at the $P \leq 0.01$ level.

cointegrated with real income. Further, as Table 11 shows, the estimated interest semielasticities are all economically sensible as well as statistically significant (based on the relevant χ^2 test). Adding observations from the 1980's, however, reduces the significance of the finding in favor of cointegration to the 0.10 level for the base and eliminates it altogether for M1 and credit. (Here, using M2 does result in a rejection at the 0.10 level.) For the 1970–1990 period, there is no evidence of cointegration in this form for any of the four aggregates.[24]

It is always possible, of course, that these results are subject to distortion because of the imposition of a unit income elasticity. To check this possibility, Table 10C presents analogous test statistics based on the cointegrating equation

$$(7) \qquad (m_t - p_t) - \beta x_t - \gamma r_t = e_t.$$

With income elasticity β also free to be estimated from the data, the test statistic warrants rejecting the no-cointegration null hypothesis at the 0.05 level or better for each of the four aggregates in the 1960–1979 sample. (In the context of comparisons to the results shown in Table 10A, the estimated β value for M2 here is only 0.6, and the $\beta = 1$ restriction is rejected at the 0.01 level.)[25] Once again, however, merely adding observations from the 1980's eliminates these positive results except for M2, for which the test statistic remains significant at the 0.10 level. For the 1970–1990 sample, none of the four aggregates is significantly cointegrated with income, even at the 0.10 level.

Finally, at least in principle it is also possible that for broad aggregates like M2 and credit, for which sensitivity to interest rates is presumably small, including the interest rate in the cointegrating vector reduces the power of the test of the no-cointegration null hypothesis against the alternative that the aggregate and income are cointegrated. Indeed, the familiar injunction to the central bank to set any given year's money growth target so as to offset any error in achieving the previous year's target, thereby eliminating any base drift, implicitly rests on the assumption not only that money and income are cointegrated but also that the cointegrating vector does not include an interest rate. (If the cointegrating vector does include an interest rate, it would be appropriate to offset only that part of the prior year's difference between actual and targeted money growth not due to permanent changes in the interest rate.)

As a check against the possibility that the negative results reported in Table 10B and

[24] William Poole (1988), among others, has advocated the use of a long-term rather than short-term interest rate in the money-demand function. If the Aaa corporate bond rate is used for r in (5), the resulting test statistic is significant at the 0.01 level for M2 and at the 0.10 level for M1 (but not even at the 0.10 level for either of the other two aggregates) for the 1960–1979 sample and significant at the 0.10 level for M2 (but not even at that level for any of the other three aggregates) for the 1960–1990 sample. The question of which interest rate to use is central to much other work on cointegration. Stock and Watson (1991), for example, experimented with a smoothed version of the Treasury bill rate (which, of course, looks more like a bond rate).

[25] By contrast, freeing β to be estimated casts doubt on the evidence for M1; here, the estimated values of β (−2.8) and γ (0.414) are both economically senseless. Given that Lucas (1988) was working with M1 data, these results provide an additional perspective on his rationale for imposing $\beta = 1$. (Stock and Watson [1991] similarly found that in trivariate systems including real income, money, and an interest rate, estimation over postwar data results in a bimodal Johansen likelihood surface, with one peak corresponding to economically sensible parameter values and the other corresponding to absurd values.)

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C might be due to the inclusion of the interest rate, Table 10D therefore presents Johansen test statistics based on the cointegrating equation

$$(8) \qquad (m_t - p_t) - \beta x_t = e_t.$$

With the interest rate excluded, M2 and credit are the only aggregates for which there is any significant evidence of cointegration with income in the 1960–1979 sample. Extending the sample through 1990 reduces this significance for M2 to the 0.10 level and eliminates it altogether for credit. For the 1970–1990 sample, there is again no evidence of cointegration for any of the four aggregates.

In sum, whatever the situation may have been before the 1980's, it is no longer possible to discern from the data a stable long-run relationship between income and the monetary base, M1, or credit, either with or without allowance for the effect of interest rates, and the evidence of such stability in the case of M2 strictly depends on the inclusion of data from the 1960's.[26] To be sure, because the statistical substance of these results is the failure to reject the null hypothesis of no cointegration, they do not (and cannot) prove that no such relationship now exists.[27] They do show, however, that the evidence in favor of the kind of long-run stability of the money–income relationship that cointegration represents has become weaker over time, so much so that any presumption in favor of such a relationship must reflect prior beliefs, rather than the evidence contained in the data now available. If for some reason the central bank happens to be conducting monetary policy by using any of these aggregates as an intermediate target, the results therefore show that the data no longer provide a compelling empirical basis for the injunction to resist in full the base drift that arises from past failures to achieve the targeted growth rate.

IV. Summary of Conclusions and Policy Implications

The empirical findings presented in this paper consistently show that evidence based on the most recent U.S. experience does not indicate a close or reliable relationship between money and nonfinancial economic activity. Merely extending the analysis to include data from the 1980's sharply weakens the time-series evidence from prior periods showing that such relationships existed between money and nominal income or between money and either real income or prices considered separately. Focusing on data from 1970 onward destroys this evidence altogether. The deterioration of the evidence supporting a relationship to either real or nominal income, or to prices, appears not just for M1 but for other monetary aggregates and for credit as well. Evidence from earlier samples indicating cointegration of real income and real money balances (with due allowance for the effect of interest rates) also disintegrates when the sample extends into the 1980's.

These changes over time in observable empirical relationships bear strongly negative implications for many familiar monetary-policy frameworks that center the design and implementation of policy on money (or credit). The point is not just that the money–income relationship does not satisfy the stringent conditions that would be required to render optimal the strict use of

[26] Substituting domestic absorption in place of income, an idea sometimes suggested in the empirical money-demand literature, makes no material difference for the results reported here.

[27] Moreover, as Engle and Granger (1987) demonstrated in the case of the augmented Dickey-Fuller test, tests of the null hypothesis of no cointegration tend to display low power against alternatives corresponding to cointegration. For example, if disturbance e in the cointegrating equation is stationary but autocorrelated with first-order coefficient 0.9 and the sample is of size 100, Engle and Granger showed that the Dickey-Fuller test would, on average, reject the "no cointegration" null hypothesis at the 0.05 level only 15 percent of the time. (Corresponding test results for the Johansen statistic are not available.) However, for the purposes of monetary policy (including in particular the question of what to do about base drift), it matters little whether shocks to the money–income relationship are truly permanent or only very highly persistent. First-order autocorrelations of e equal to 0.95 or 0.99, for example, still imply shocks with half-lives equal to 13.5 quarters or 69 quarters, respectively.

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money as an intermediate target. More importantly, there is no evidence to show that fluctuations in money contain any information about subsequent movements in income or prices. Further, even if the central bank did choose for some reason to base monetary policy on the use of money as an intermediate target, in the absence of evidence indicating that money and income are cointegrated there would be no empirical grounds on which to resist base drift.

The paper's one finding with potentially positive implications for monetary policy is that the spread between the commercial paper rate and the Treasury bill rate has contained information about subsequent movements in real income, information that is highly significant regardless of the sample period under investigation. It is difficult to imagine that the Federal Reserve System could use this interest rate spread as an intermediate policy target, and it is doubtful that the relationships found here would continue to hold if it did. By contrast, in a world in which previously standard financial quantities can no longer serve as policy information variables because they no longer contain information about the macroeconomic outcomes that monetary policy seeks to affect, policymakers must somehow fill the resulting vacuum by exploiting information from wherever they happen to find it.

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